

Topic 1.11 - Equivalent Representations of Polynomial and Rational Expressions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

The same polynomial or rational expression can be written in several useful forms. Expanded form emphasizes degree and leading behavior; factored form reveals zeros and multiplicities; quotient-remainder form reveals division structure and long-run behavior. Algebraic equivalence must be justified, and canceled factors never erase original domain restrictions.

Learning targets

- Rewrite polynomial expressions in factored, expanded, and quotient-remainder forms and explain what each form reveals.
- Use polynomial division and the Remainder and Factor Theorems to make and justify conclusions.
- Rewrite rational expressions while preserving exclusions, holes, zeros, and asymptotes.

Language and structure

Term	Meaning in this topic
equivalent expressions	Expressions with the same value at every input common to their domains.
factor form	A product form that makes zeros and multiplicities visible.
quotient-remainder form	$P(x) = D(x)Q(x) + R(x)$, where the degree of R is less than the degree of D .
Remainder Theorem	The remainder when $P(x)$ is divided by $x - c$ equals $P(c)$.
domain restriction	An input excluded by the original expression, even if a factor later cancels.

Launch - reason before calculating

The expressions $x^3 - x^2 - 8x + 12$ and $(x - 2)^2(x + 3)$ look different. Without graphing, list one feature that is easier to read from each form and explain why both forms should be retained in a complete analysis.

Concept 1: Selecting a form that reveals structure

Core idea

Factoring and expanding are reversible when performed correctly. Use factored form for zeros, multiplicity, sign, and cancellation; expanded form for degree, leading coefficient, coefficient comparison, and some regression models. A complete solution often moves between forms rather than treating one as universally best.

Worked example - annotate the reasoning

Problem. Rewrite $P(x) = x^3 - 4x^2 - x + 4$ in factored form and state what each representation reveals.

Model reasoning. Group terms: $x^2(x - 4) - 1(x - 4) = (x - 4)(x^2 - 1) = (x - 4)(x - 1)(x + 1)$. Expanded form reveals degree 3 and positive leading coefficient. Factored form reveals simple zeros $-1, 1, 4$.

Check your understanding

Independent

No calculator

Factor $x^4 - 5x^2 + 4$ completely over the reals and identify all real zeros with multiplicity.

Concept 2: Division, remainders, and factor tests

Core idea

Polynomial division produces an identity valid for every input: $P = DQ + R$. For a linear divisor $x - c$, the remainder is the number $P(c)$. Thus $x - c$ is a factor exactly when $P(c) = 0$. Synthetic division is efficient, but the quotient-remainder identity should still be stated.

Worked example - annotate the reasoning

Problem. Divide $2x^3 - 3x^2 + 4x - 5$ by $x - 2$ and verify the remainder independently.

Model reasoning. Synthetic division with 2 gives quotient $2x^2 + x + 6$ and remainder 7. Therefore $P = (x - 2)(2x^2 + x + 6) + 7$. Also $P(2) = 16 - 12 + 8 - 5 = 7$.

Check your understanding

Independent

No calculator

Divide $x^4 - 3x^3 + 2x^2 + 5x - 7$ by $x + 1$ and interpret the remainder.

Concept 3: Rational equivalence and preserved restrictions

Core idea

Canceling a common factor changes the written expression but not the original domain. The simplified expression describes the graph except at canceled inputs. A canceled denominator factor creates a hole; an uncanceled denominator factor can create a vertical asymptote. State exclusions before canceling.

Worked example - annotate the reasoning

Problem. Simplify $\frac{x^2-9}{x^2-x-6}$ and identify every discontinuity.

Model reasoning. The original denominator factors as $(x-3)(x+2)$, so $x=3, -2$ are excluded. Canceling $x-3$ gives $(x+3)/(x+2)$. There is a hole at $(3, 6/5)$ and a vertical asymptote at $x=-2$.

Check your understanding

Independent

No calculator

Simplify $\frac{x^2-4x+4}{x^2-4}$ and identify the zero, hole, and vertical asymptote.

Representation laboratory

Connect at least two representations

For $P(x) = (x-2)^2(x+3) = x^3 - x^2 - 8x + 12$, create a two-column feature inventory: features visible immediately in factored form and features visible immediately in expanded form. Then identify one feature requiring additional reasoning in either form.

Challenge laboratory

Challenge - justify every claim

Expand $(2x - 3)^4$ using structure rather than repeated multiplication. Then explain how the coefficient pattern is connected to Pascal's triangle and verify the constant and leading terms without completing the full expansion again.

Error clinic and calculator strategy

Common errors to diagnose

- Canceling terms across addition, such as changing $(x + 3)/x$ into 3.
- Declaring two rational functions identical without noting an excluded input in one domain.
- Using synthetic division with the wrong sign for a divisor such as $x + 4$.
- Reporting a factor from an approximate zero without checking the remainder.

Technology decision

A computer algebra system can factor, expand, and divide, but it may suppress original restrictions or return a form that hides the requested feature. Record the original domain first, and verify a claimed factor with substitution or multiplication.

AP-style synthesis**Multi-part reasoning task**

Let $P(x) = x^4 - 5x^3 + 2x^2 + 20x - 24$.

- A. Verify that $P(x) = (x - 3)(x - 2)^2(x + 2)$ by a reversible algebraic method.

- B. Use the most informative form to identify all zeros and graph behavior at each zero.

- C. Rewrite $\frac{P(x)}{(x-2)(x+2)}$ in simplest form while preserving all original restrictions, and identify every hole.

Exit ticket**Before you leave**

1. What does the remainder on division by $x - c$ equal?

2. Why can canceling a factor create a hole rather than remove a restriction?

3. Which form is usually best for identifying multiplicity?
